

Inline and Display Mathematics

Source-backed grouped formula sample

1 Expressions

Expression 1. The following expression is taken from a source-backed formula corpus.

$$T_{rh} \approx \left(\frac{90}{\pi^2 g_*} \right)^{\frac{1}{4}} \min \left(\sqrt{H(\phi_e)}, \sqrt{\Gamma} \right) \approx \left(\frac{30}{\pi^2 g_*} \right)^{\frac{1}{4}} \min \left(\Delta, \left(\frac{3}{8\pi^2} \right)^{\frac{1}{4}} p^{\frac{3}{2}} \kappa^{\frac{5}{2p}} \Delta^{3 - \frac{5q}{2p}} \right)$$

Expression 2. The following expression is taken from a source-backed formula corpus.

$$\mathcal{L} = \frac{1}{2} \left\{ \partial_m \phi_i \partial^m \phi_i + \bar{\psi}_i \gamma^m \partial_m \psi_i - \frac{\partial \mathcal{W}}{\partial \phi_i} \frac{\partial \mathcal{W}}{\partial \phi_i} - \frac{\partial^2 \mathcal{W}}{\partial \phi_i \partial \phi_j} \bar{\psi}_i \psi_j \right\}.$$

Expression 3. The following expression is taken from a source-backed formula corpus.

$$\partial_\varphi \left\{ \partial_z S^z \partial_{\bar{z}} S^{\bar{z}} - \partial_z S^{\bar{z}} \partial_{\bar{z}} S^z + \mathcal{H} \right\} = \partial_z p + \partial_{\bar{z}} \bar{p} + \partial_\varphi \mathcal{H} = \partial_\varphi \left\{ \partial_z S + \partial_{\bar{z}} \bar{S} + \mathcal{H} \right\} = 0.$$

Expression 4. The following expression is taken from a source-backed formula corpus.

$$I_{\psi, \tilde{\psi}} \equiv \frac{1}{2} \int_M \epsilon^{\mu\nu\rho\sigma} \left[\tilde{\psi}^{A'}_{\mu} e_{AA'\nu} D_\rho \psi^A_\sigma - \psi^A_\mu e_{AA'\nu} D_\rho \tilde{\psi}^{A'}_\sigma \right] d^4x,$$

Expression 5. The following expression is taken from a source-backed formula corpus.

$$S = -\frac{1}{2} \int d\sigma d\tau [\partial_\mu \chi_1 \partial^\mu \chi_1 + 2\chi_2 \partial_\mu \chi_1 \partial^\mu \chi_3 + \frac{1}{1 + \chi_1^2} \partial_\mu \chi_2 \partial^\mu \chi_2 + (\chi_2^2 + \frac{1}{1 + \chi_1^2}) \partial_\mu \chi_3 \partial^\mu \chi_3 + \partial_\mu t \partial^\mu t].$$